HOF { map, filter } & loops { for of & for in loop } lab Questions : By Gagan Baghel

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🔷 for...of Loop (10 Questions)

1. **Use `for...of` to print all elements of an array `[1, 2, 3, 4]`.**

   Instructions:

   - Create an array.
   - Write a `for...of` loop.
   - Inside the loop, `console.log` each element.

2. **Use `for...of` to print each character of the string `"hello"`.**

   Instructions:

   - Create a string variable.
   - Use `for...of` because strings are iterable.
   - Print each character.

3. **Use `for...of` to print all numbers in `[10, 20, 30]`.**

   Instructions:

   - Define the array.
   - Loop using `for...of`.
   - Print each number.

4. **Use `for...of` to find the sum of numbers in `[5, 10, 15]`.**

   Instructions:

   - Make a `sum = 0` variable.
   - Loop using `for...of`.

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- Add each number to `sum`.
- Print the final sum.

5. **Use `for...of` to print all colors in `["red", "green", "blue"]`.**

   Instructions:

   - Define the colors array.
   - Use `for...of`.
   - Print each color.

6. **Use `for...of` to print the square of each number in `[2, 4, 6]`.**

   Instructions:

   - Create the array.
   - Loop using `for...of`.
   - Inside loop, print `number * number`.

7. **Use `for...of` to print names in `["Alice", "Bob", "Charlie"]`.**

   Instructions:

   - Create the array.
   - Use `for...of`.
   - Print each name.

8. **Use `for...of` to print each element of `[true, false, true]`.**

   Instructions:

   - Create the array.
   - Loop.
   - Print each boolean value.

9. **Use `for...of` to count how many elements are in `[1, 2, 3, 4, 5]`.**

   Instructions:

   - Make a `count = 0`.
   - For each element, increment count.

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- Print count.

10. **Use `for...of` to print the doubled value of each number in `[1, 3, 5]`.**

    Instructions:

    - Create the array.
    - Loop using `for...of`.
    - Print `num * 2`.

🔷 for...in Loop (10 Questions)

1. **Use `for...in` to print all keys of `{a: 1, b: 2, c: 3}`.**

    Instructions:

    - Create the object.
    - Use `for...in` (it gives keys).
    - Print each key.

2. **Use `for...in` to print all values of `{x: 10, y: 20}`.**

    Instructions:

    - Create the object.
    - Loop using `for...in`.
    - Use `obj[key]` to print values.

3. **Use `for...in` to print both keys and values of `{name: "John", age: 25}`.**

    Instructions:

    - Create the object.
    - Loop using `for...in`.
    - Print key and value together.

4. **Use `for...in` to count the number of keys in `{p: 5, q: 10, r: 15}`.**

    Instructions:

    - Create a counter.

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- For each key, increment counter.
- Print final counter.

5. **Use `for...in` to print each property of `{fruit: "apple", color: "red"}`.**

   Instructions:

   - Create object.
   - Loop using `for...in`.
   - Print both key and value.

6. **Use `for...in` to create a string of keys from `{a: 1, b: 2}`.**

   Instructions:

   - Create empty string.
   - Loop through keys.
   - Add each key to string.
   - Print the string.

7. **Use `for...in` to check if the key `"age"` exists in `{name: "Sam", age: 30}`.**

   Instructions:

   - Create the object.
   - Loop keys using `for...in`.
   - Check if key is `"age"`.
   - Print true if exists.

8. **Use `for...in` to print values greater than 10 in `{a: 5, b: 15, c: 25}`.**

   Instructions:

   - Loop over keys.
   - Check if value > 10.
   - Print value.

9. **Use `for...in` to print the type of each value in `{name: "Eva", age: 22}`.**

   Instructions:

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- Loop keys.
- Print `typeof obj[key]`.

10. **Use `for...in` to copy all keys into an array from `{id: 1, code: 2}`.**

    Instructions:

    - Create empty array.
    - Loop keys using `for...in`.
    - Push each key into array.
    - Print array.

🔷 filter (10 Questions)

1. **Use `filter` to get even numbers from `[1, 2, 3, 4, 5, 6]`.**

   Instructions:

   - Create array.
   - Use `filter`.
   - Condition: `number % 2 === 0`.

2. **Use `filter` to get odd numbers from `[10, 11, 12, 13]`.**

   Instructions:

   - Use `filter`.
   - Condition: `number % 2 !== 0`.

3. **Use `filter` to get numbers greater than 5 from `[2, 8, 3, 10]`.**

   Instructions:

   - Use `filter`.
   - Condition: `number > 5`.

4. **Use `filter` to get words with length greater than 3 from `["hi", "hello", "to", "world"]`.**

   Instructions:

   - Use `filter`.

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- Condition: `word.length > 3`.

5. **Use `filter` to get numbers less than 10 from `[5, 15, 8, 20]`.**

   Instructions:

   - Use `filter`.
   - Condition: `number < 10`.

6. **Use `filter` to get strings starting with `"a"` from `["apple", "banana", "avocado"]`.**

   Instructions:

   - Use `filter`.
   - Condition: `word.startsWith("a")`.

7. **Use `filter` to get positive numbers from `[-3, -1, 0, 2, 4]`.**

   Instructions:

   - Use `filter`.
   - Condition: `number > 0`.

8. **Use `filter` to get values equal to `"cat"` from `["dog", "cat", "lion", "cat"]`.**

   Instructions:

   - Use `filter`.
   - Condition: `value === "cat"`.

9. **Use `filter` to get numbers divisible by 3 from `[3, 6, 7, 9, 10]`.**

   Instructions:

   - Use `filter`.
   - Condition: `number % 3 === 0`.

10. **Use `filter` to get names with length exactly 4 from `["John", "Paul", "Sam"]`.**

    Instructions:

    - Use `filter`.
    - Condition: `name.length === 4`.

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🔷 map (10 Questions)

1. **Use `map` to double each number in `[1, 2, 3]`.**

   Instructions:

   - Create array.
   - Use `map`.
   - Return `num * 2`.

2. **Use `map` to square each number in `[2, 3, 4]`.**

   Instructions:

   - Use `map`.
   - Return `num * num`.

3. **Use `map` to convert each string in `["a", "b", "c"]` to uppercase.**

   Instructions:

   - Use `map`.
   - Return `str.toUpperCase()`.

4. **Use `map` to get the length of each word in `["hi", "hello", "world"]`.**

   Instructions:

   - Use `map`.
   - Return `word.length`.

5. **Use `map` to add 10 to each number in `[5, 10, 15]`.**

   Instructions:

   - Use `map`.
   - Return `num + 10`.

6. **Use `map` to change numbers into strings from `[1, 2, 3]`.**

   Instructions:

   - Use `map`.
   - Return `String(num)` or `num.toString()`.

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7. **Use `map` to get the first character of each word in `["cat", "dog", "lion"]`.**

   Instructions:

   - Use `map`.
   - Return `word[0]`.

8. **Use `map` to make each number negative from `[1, 2, 3]`.**

   Instructions:

   - Use `map`.
   - Return `num`.

9. **Use `map` to create an array of `true` / `false` if numbers are even in `[2, 5, 8]`.**

   Instructions:

   - Use `map`.
   - Return `num % 2 === 0`.

10. **Use `map` to append `"!"` to each word in `["hi", "bye"]`.**

    Instructions:

    - Use `map`.
    - Return `word + "!"`.

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